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A consolidated record of internship project work, covering front-end e-commerce development, data analytics with Power BI, and full-stack application development with AI integration.

01

Amazon : The Shopping King
— E-Commerce Web Platform

02

Hospital ER Dashboard —
Power BI Analytics

03

College Sahayak — Full-Stack
Web Application

INTERN

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ROLE

Software
Development
Intern

ORGANIZATION

Somayu
Infotech

DURATION

June
2026 –
August
2026

MENTOR

Mrs.
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DATE

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August
2026

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01

PROJECT ONE

Amazon : The Shopping King E-Commerce Web Platform

A modern, responsive e-commerce web platform built with a dark premium aesthetic, featuring dynamic categories, hot deals product catalog, cart interaction, user authentication, and profile management.

HTML5

CSS3 / Inter Font

JavaScript (ES6)

Firebase Auth

Glassmorphism UI

PROJECT 01 OF 03

Amazon : The Shopping King

CATEGORY	TYPE	DURATION	STATUS
Front-End E-Commerce	Multi-Page Web Platform	2 Weeks	Completed & Redesigned

Overview

Amazon : The Shopping King is a comprehensive, multi-page front-end e-commerce platform designed to deliver an intuitive and visually striking shopping experience. Featuring a modern dark-themed aesthetic with vibrant orange accents, the site showcases a curated product catalog across multiple categories, interactive deal banners, smooth scroll animations, and user account flows.

Key Features

- Responsive Navigation & Glassmorphism Navbar:** Sticky, blur-effect header with brand identity, search navigation, and mobile hamburger menu.
- Dynamic Hero & Hot Deals Section:** Animated floating badges, real-time product cards with discount tags, star ratings, and interactive "Add to Cart" state management.
- Category Discovery:** 6 core categories (Electronics, Fashion, Home & Living, Sports, Books, Beauty) with custom hover gradient accents.
- User Management & Privacy:** Complete account portal containing login forms, registration, user profile, contact desk, and privacy policy sections.
- Firebase Integration:** Firebase authentication setup for secure sign-in and user state persistence.

Technology Stack

HTML5

CSS3 (Modern Variables & Flexbox/Grid)

JavaScript (ES6 Modules)

Firebase Auth

Google Fonts (Inter)

Unsplash API

LIVE DEPLOYMENT (VERCEL)

<https://amazon-e-shop.vercel.app>

INTERFACE SHOWCASE

Desktop View — Homepage

Full-page capture of the redesigned Amazon : The Shopping King homepage on a desktop viewport, showcasing the hero banner, stats bar, category cards, and deal showcase.

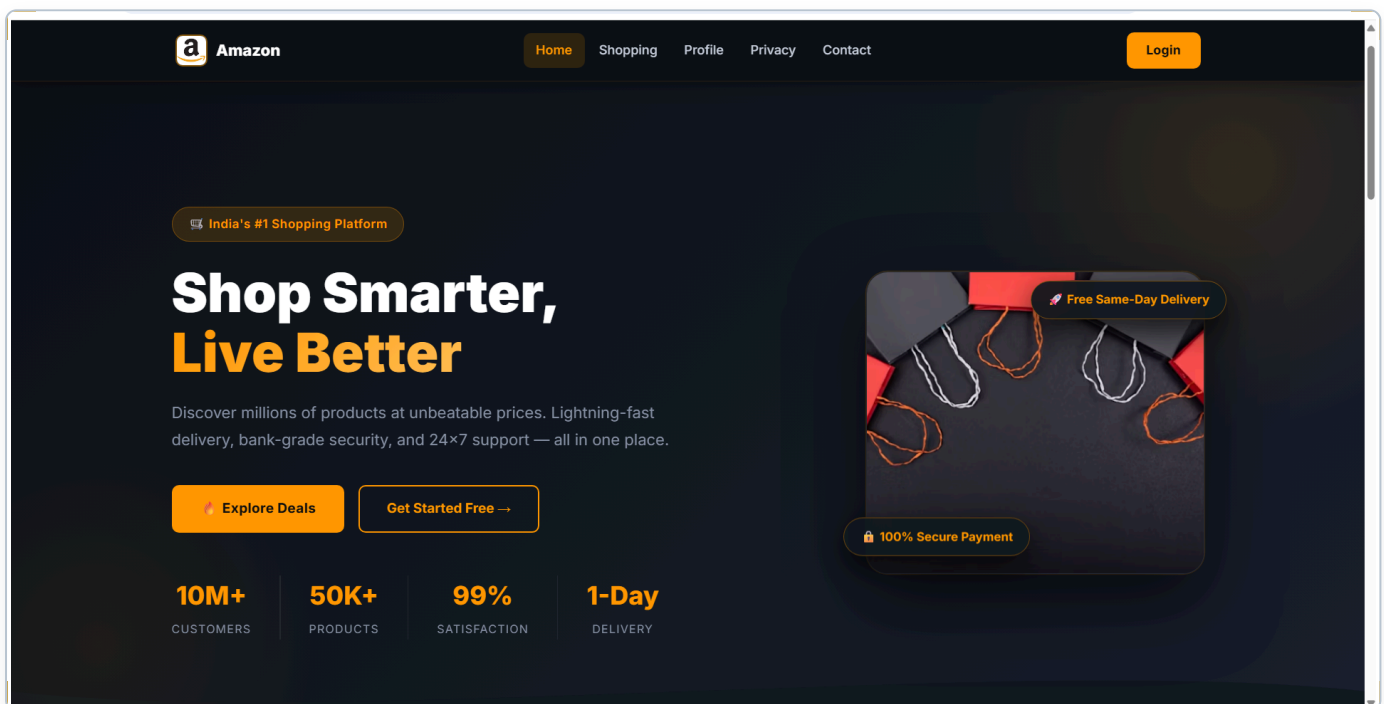


Fig. 1.1 — Homepage, Dark E-Commerce UI Theme

INTERFACE SHOWCASE

Additional Views

Supporting screens covering key modules of the e-commerce platform — product listing, user authentication, customer profile, and contact support.

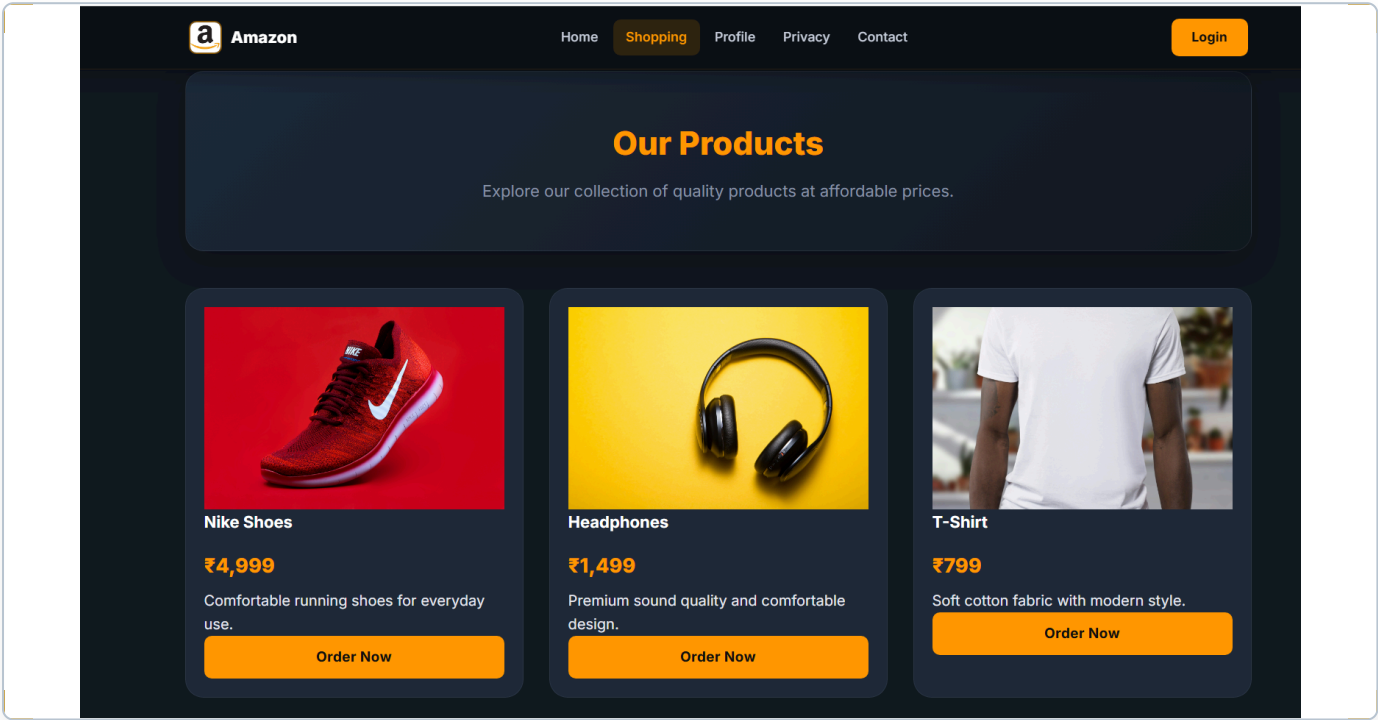


Fig. 1.2 — Shopping Catalog Page

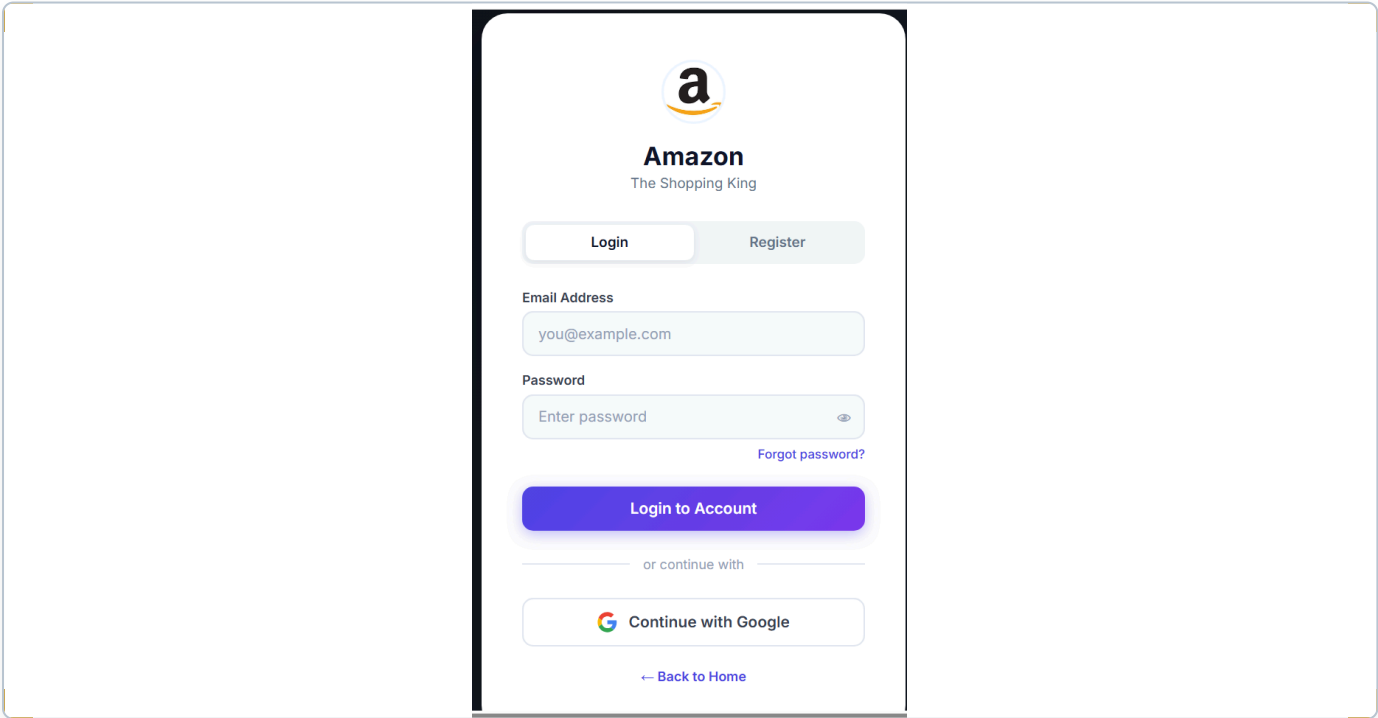


Fig. 1.3 — Login & Authentication Portal

02

PROJECT TWO

Hospital ER Dashboard Power BI Analytics

An interactive analytics dashboard for Hospital Emergency Room data with 9,200+ patient records — providing real-time insights into wait times, satisfaction scores, demographics, and department referrals.

Power BI

Power Query

DAX

Data Modeling

CSV / Excel

PROJECT 02 OF 03

Hospital ER Dashboard — Power BI

CATEGORY	TOOL	DATASET	STATUS
Data Analytics & BI	Microsoft Power BI	9,200+ Patient Records	Completed

Overview

This project involved building an interactive analytics dashboard for a Hospital Emergency Room dataset containing over 9,200 patient records. The dashboard provides real-time insights into patient demographics, wait times, satisfaction scores, department referrals, and admission patterns — enabling hospital administrators to identify bottlenecks and improve patient care efficiency.

Methodology

- 1

Data Collection & Cleaning
Imported the Hospital ER CSV dataset (9,217 rows × 12 columns) into Power Query. Cleaned inconsistencies in date formats, handled missing satisfaction scores, and standardized patient race and gender categories.
- 2

Data Modeling
Structured the dataset with calculated columns for age groups, wait-time brackets, and admission flags. Defined relationships for cross-filtering across demographics, departments, and time periods.
- 3

DAX Measures & KPIs
Built calculated measures for average wait time, patient satisfaction score, admission rate percentage, and department-wise referral counts. Created KPI cards for at-a-glance performance monitoring.
- 4

Dashboard Design
Assembled an interactive report with slicers for gender, age group, and department. Included trend charts, bar graphs, KPI cards, and drill-down capabilities for detailed analysis.

Tools & Techniques

- Power BI Desktop
- Power Query
- DAX
- Data Modeling
- CSV / Excel
- KPI Design

DASHBOARD SHOWCASE

Monthly View Dashboard

Monthly view of the Hospital ER Dashboard showing KPI cards for total patients (4,878), average wait time (35.5 min), satisfaction score (4.96), and patients referred (1,992). Includes age group distribution, gender split, department referral breakdown, admission status, patients seen within 30-minute target, and a day-hour heatmap for staffing optimization.

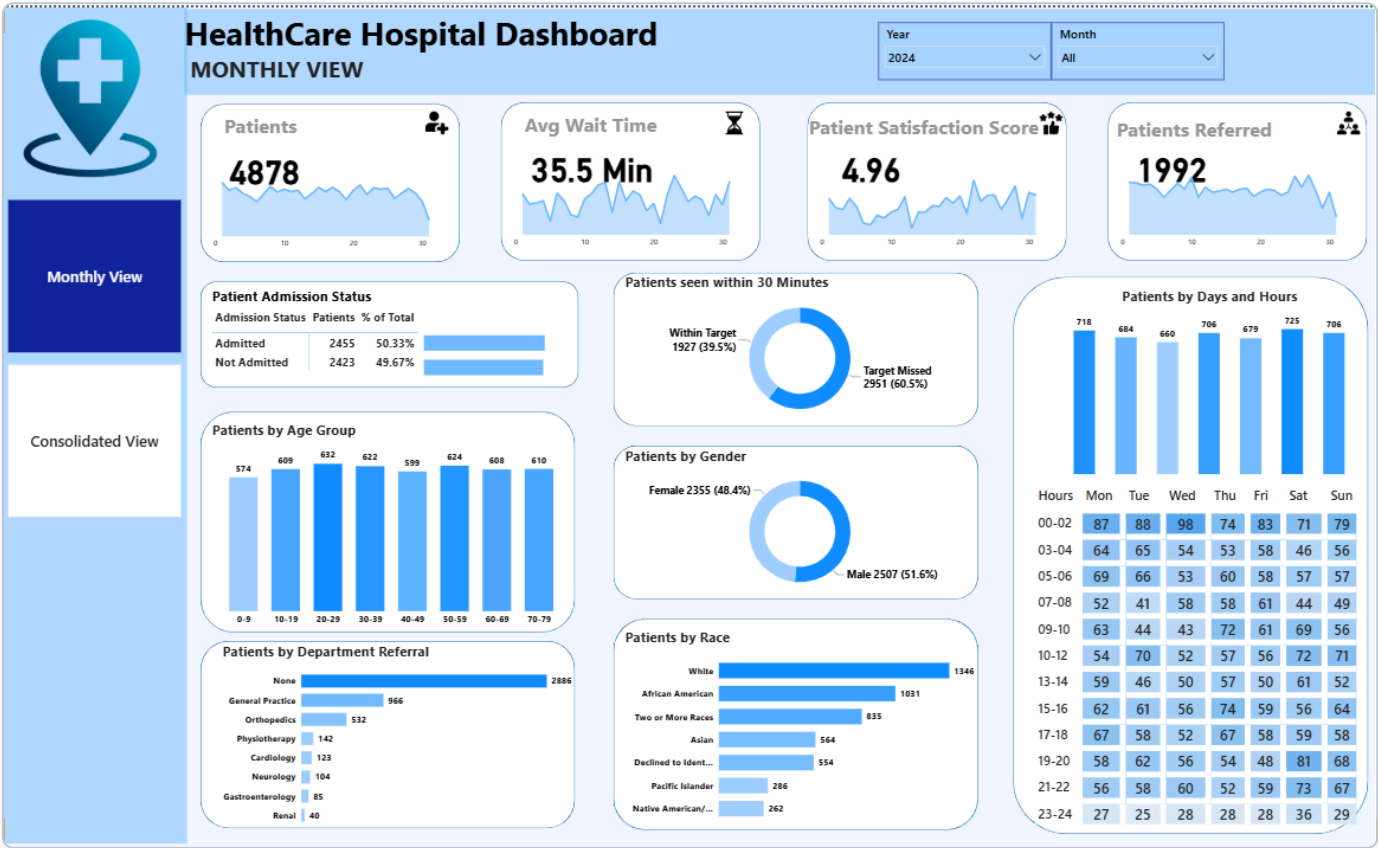


Fig. 2.1 — Power BI Dashboard, Monthly View (Year 2024)

DASHBOARD SHOWCASE

Consolidated View Dashboard

Consolidated view across the full date range (April 2023 – October 2024) showing aggregate KPIs: 9,216 total patients, 35.3 min average wait time, 4.99 satisfaction score, and 3,816 referrals. This view enables trend analysis across the complete dataset for long-term strategic planning and resource allocation.

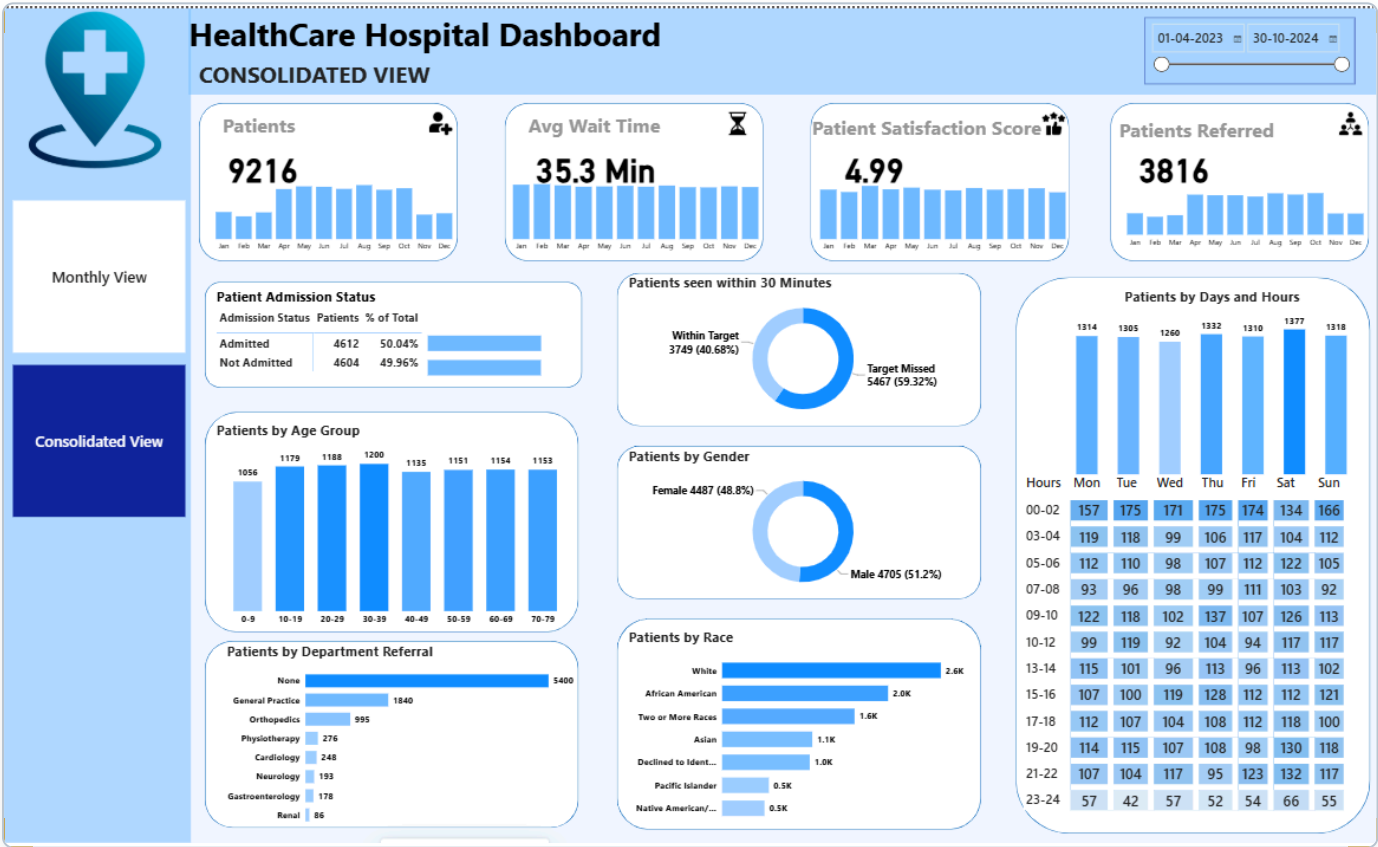


Fig. 2.2 — Power BI Dashboard, Consolidated View (Apr 2023 – Oct 2024)

03

PROJECT THREE — PRIMARY PROJECT

College Sahayak Full-Stack Web Application

A production-grade, full-stack study resource platform for MSBTE diploma students — featuring secure authentication, role-based admin panel, 10 resource categories, MSBTE curriculum browser, and an AI chatbot powered by Gemini.

React 19

TypeScript

Node.js

Express 5

Supabase

Firebase

Gemini API

Vercel

Render

PROJECT 03 OF 03 — PRIMARY PROJECT

College Sahayak — Full-Stack Application

CATEGORY	TYPE	DURATION	STATUS
Full-Stack Development	Web Application (SPA)	6 Weeks	Completed & Deployed

Overview

College Sahayak is a production-grade, full-stack web application designed to serve as a centralized study resource platform for MSBTE diploma students. The application provides organized access to study materials — lab manuals, notes, assignments, question papers, micro-projects, MSBTE important questions, lecture videos, and curriculum data — categorized by branch, semester, and subject. It features secure authentication (email/password with OTP verification + Google OAuth), role-based access control, a full admin dashboard, an integrated AI chatbot powered by the Gemini API, and is deployed on Vercel (frontend) and Render (backend) for production use.

Objectives

- Build a scalable full-stack architecture with a clean separation between React SPA client, Express API server, and Supabase (PostgreSQL) data layer.
- Implement secure authentication with email + OTP verification via Brevo, Google OAuth via Firebase, and JWT-based session management.
- Develop a role-based admin panel for managing users, materials, feedback, and application data in real time.
- Integrate a Gemini-powered AI chatbot to provide in-app study assistance for students.
- Ensure the application is fully responsive, SEO-optimized, and deployment-ready on cloud platforms.

Core Modules

- **Authentication:** Email/Password login with OTP verification, Google OAuth sync, forgot/reset password flow, and protected routes.
- **Materials Hub:** 10 resource categories (Notes, Lab Manuals, Assignments, etc.) with branch/semester/subject filtering and Google Drive integration.
- **Admin Panel:** Centralized dashboard with user management, material CRUD, feedback review, and bulk operations.
- **AI Chatbot:** Gemini API-powered conversational assistant with markdown rendering and context-aware responses.
- **MSBTE Curriculum:** Complete curriculum browser with branch, semester, and subject-level navigation.

ENGINEERING DETAILS

Technology Stack

Front End

- **React 19** with TypeScript for building a component-based, single-page interface using Vite as the build tool.
- **React Router v7** for client-side routing with protected route guards.
- Custom CSS3 styling with responsive design across all breakpoints.
- **Lucide React** for consistent iconography throughout the UI.
- **React Helmet Async** for SEO meta tags and dynamic page titles.
- **React Markdown** with GFM support for AI chatbot response rendering.

Back End

- **Node.js** with **Express 5** REST API layer serving 5 route modules (auth, users, materials, feedback, chat).
- **JWT** authentication with role-based middleware (user/admin access control).
- **Firebase Admin SDK** for Google OAuth token verification.
- **bcrypt.js** for secure password hashing.
- **Express Rate Limiter** for API abuse prevention.

Database

- **Supabase** (PostgreSQL) for persistent storage with real-time capabilities.
- Schema: users, materials, feedback, curriculum tables with role, verification, and OTP fields.

AI & Email Services

- **Google Gemini API** integration for conversational AI chatbot assistance.
- **Brevo (Sendinblue)** transactional email API for OTP delivery and verification emails.
- **Firebase Authentication** for Google OAuth sign-in flow.

Deployment & DevOps

- **Vercel** — Frontend SPA deployment with automatic CI/CD from GitHub.
- **Render** — Backend API deployment with environment variable management.
- **Git & GitHub** — Version control with feature branching.

React 19

TypeScript

Vite

Node.js

Express 5

Supabase

PostgreSQL

JWT

Firebase

Gemini API

Brevo

Vercel

Render

FRONTEND (VERCEL)

<https://internship-2k26.vercel.app>

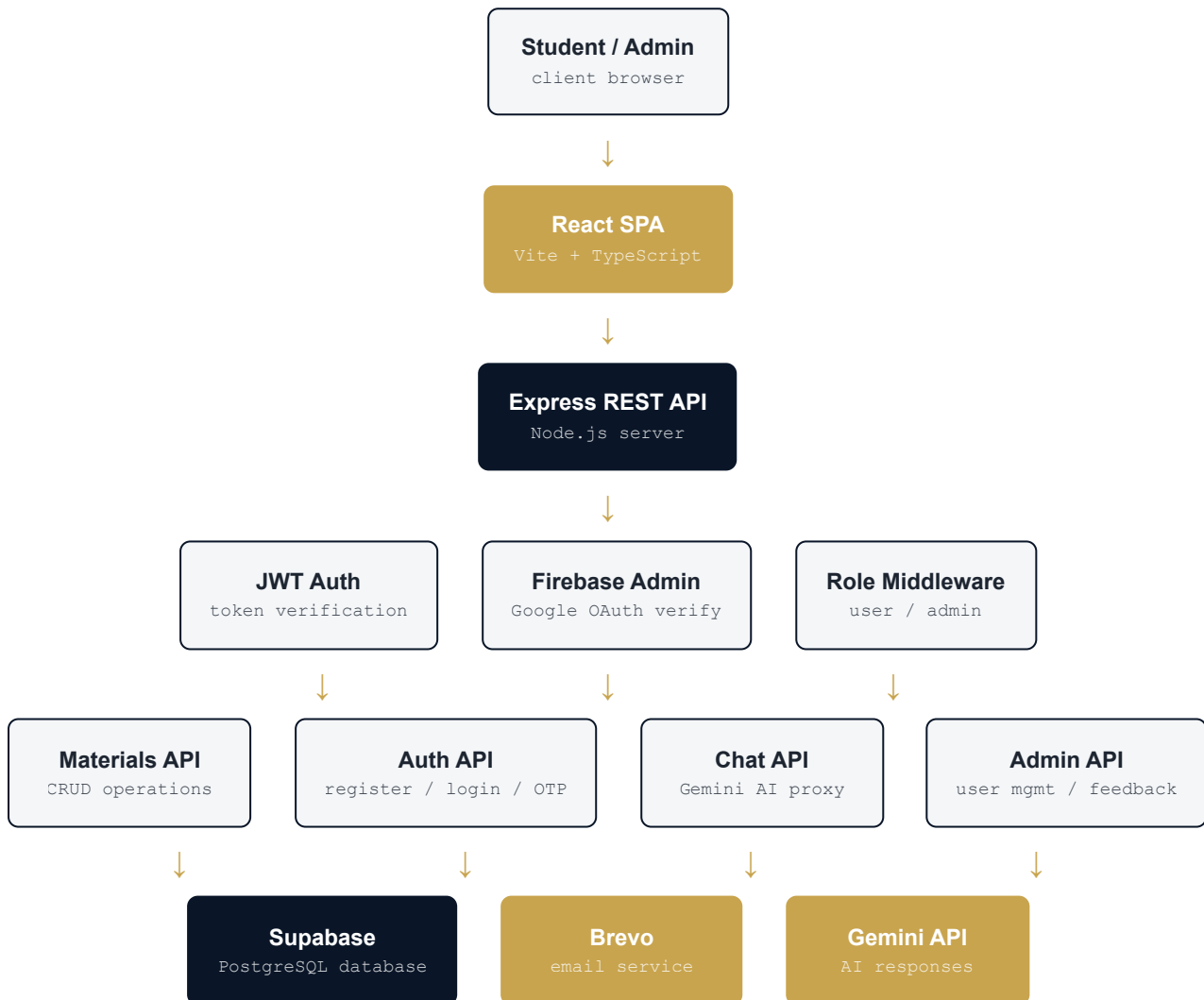
BACKEND API (RENDER)

<https://college-sahayak.onrender.com>

SYSTEM DESIGN

Architecture & Flow

High-level flow of a request through the College Sahayak application, from the client interface through authentication to the database and external services.



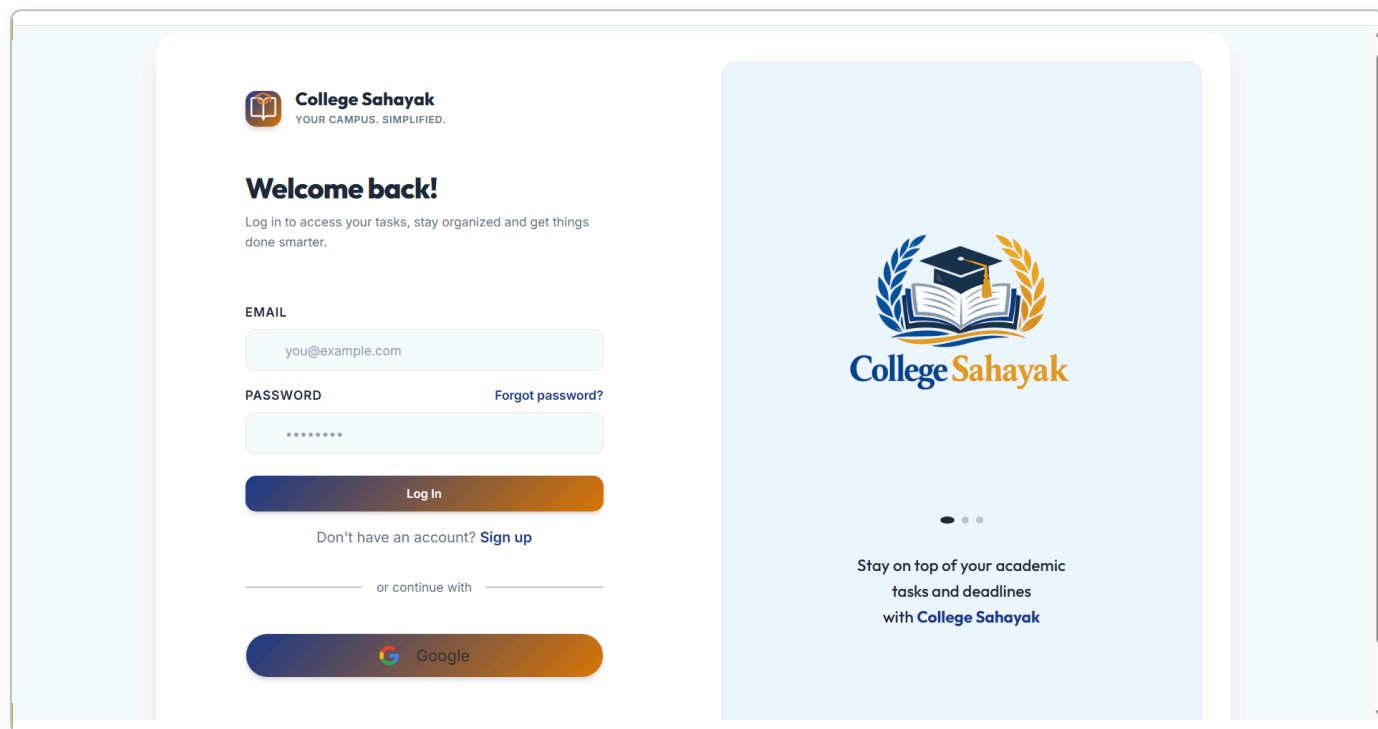
Database Schema

- **users** — id, firebase_uid, email, name, password_hash, role, is_verified, otp_code, otp_expiry, created_at
- **materials** — id, title, description, category, branch, semester, subject_code, drive_link, image_link, is_trending, created_at
- **feedback** — id, user_id, message, rating, created_at
- **curriculum** — id, branch, semester, subject_name, subject_code, type, abbreviation

INTERFACE SHOWCASE

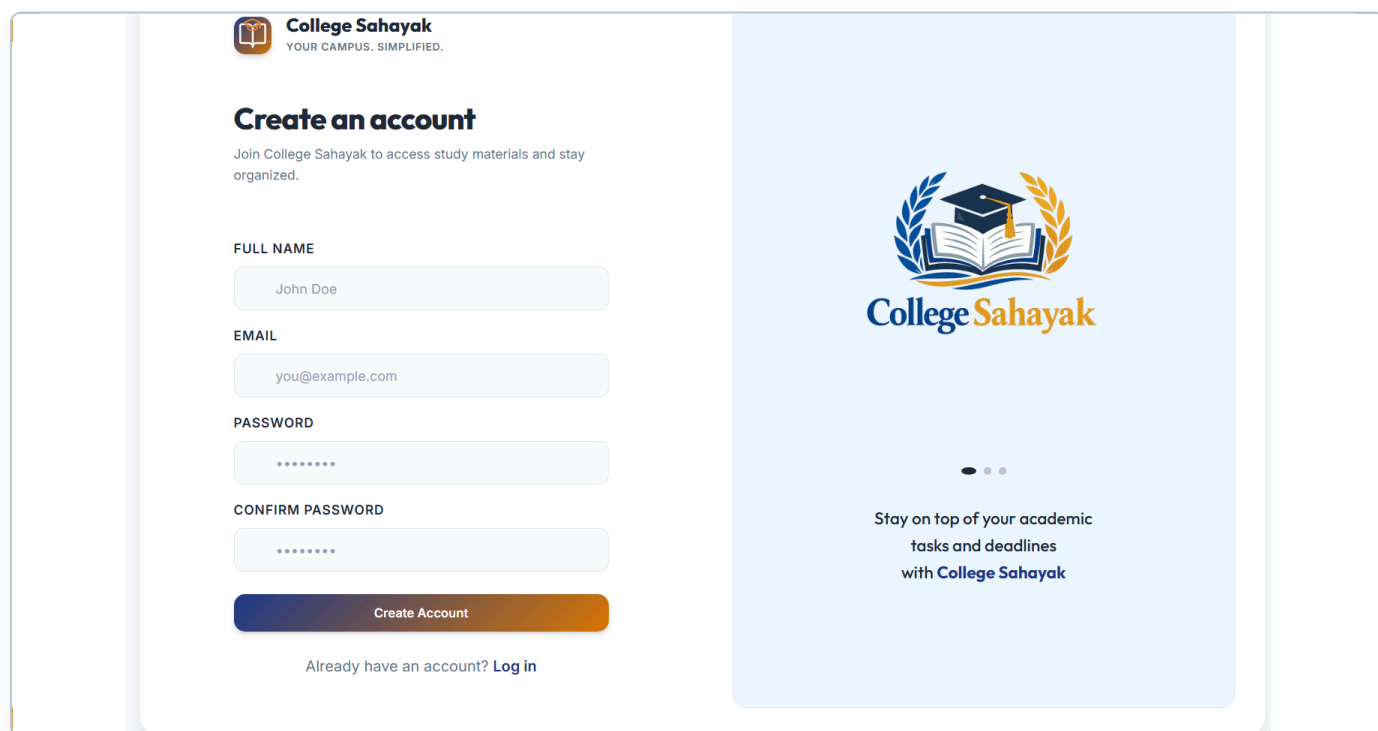
Authentication — Login & Registration

Secure authentication interface featuring both Email/Password login and Google OAuth integration, paired with user registration and form validation.



The login portal is a two-column layout. The left column contains the login form, and the right column features a large blue sidebar with the College Sahayak logo and a motivational message. The login form includes the College Sahayak logo and tagline, a 'Welcome back!' heading, a brief description of the app's benefits, and input fields for email and password. A 'Log In' button is positioned below the password field, with a 'Forgot password?' link. Below the login button is a link to 'Sign up' for users who don't have an account. At the bottom of the form is a 'or continue with' section featuring a Google OAuth button. The right sidebar contains the College Sahayak logo and the text: 'Stay on top of your academic tasks and deadlines with College Sahayak'.

Fig. 3.1 — Login Portal with Email & Google OAuth



The user registration form is a two-column layout. The left column contains the registration form, and the right column features a large blue sidebar with the College Sahayak logo and a motivational message. The registration form includes the College Sahayak logo and tagline, a 'Create an account' heading, a brief description of the app's benefits, and input fields for full name, email, password, and confirm password. A 'Create Account' button is positioned below the confirm password field. Below the button is a link to 'Log in' for users who already have an account. The right sidebar contains the College Sahayak logo and the text: 'Stay on top of your academic tasks and deadlines with College Sahayak'.

Fig. 3.2 — User Registration Form

INTERFACE SHOWCASE

Authentication — Verification & Security

Email verification using 6-digit OTP codes delivered via Brevo API, and password reset security flows.

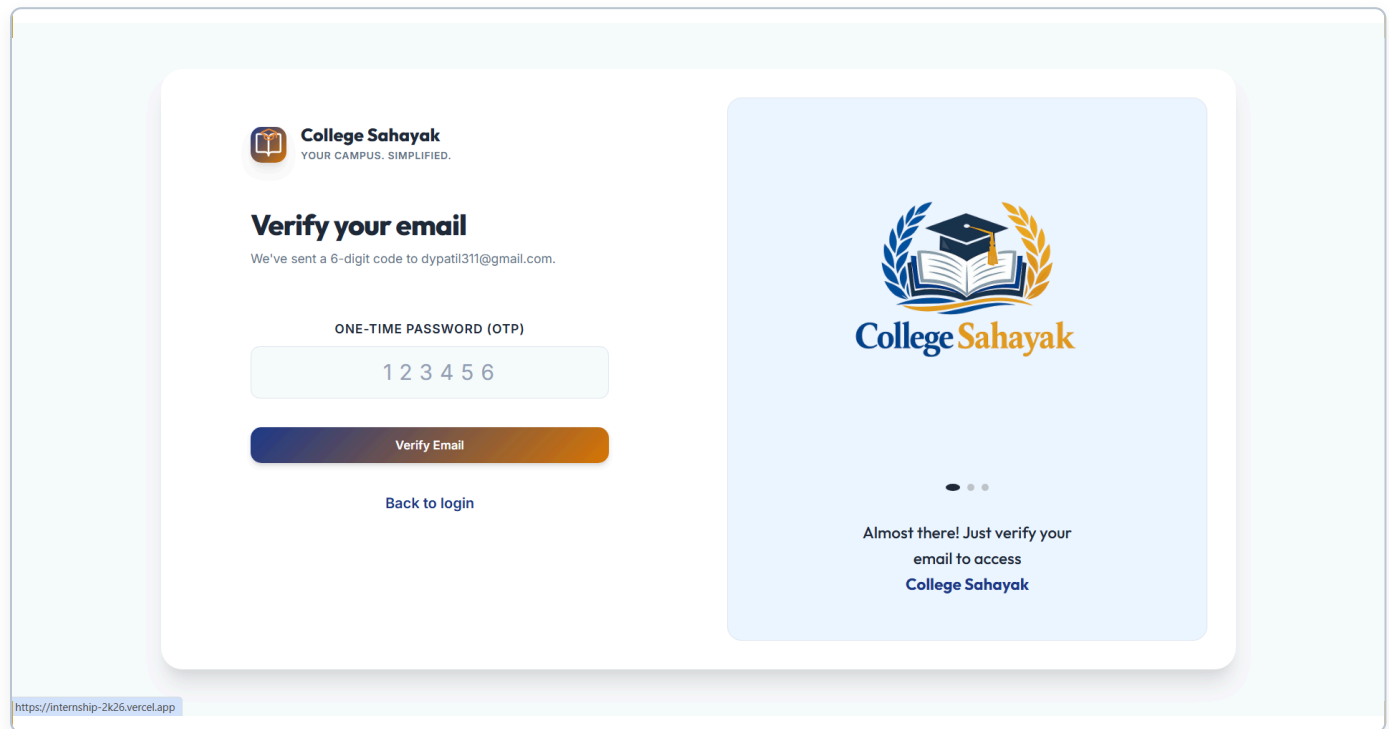


Fig. 3.3 — 6-Digit Email OTP Verification Screen

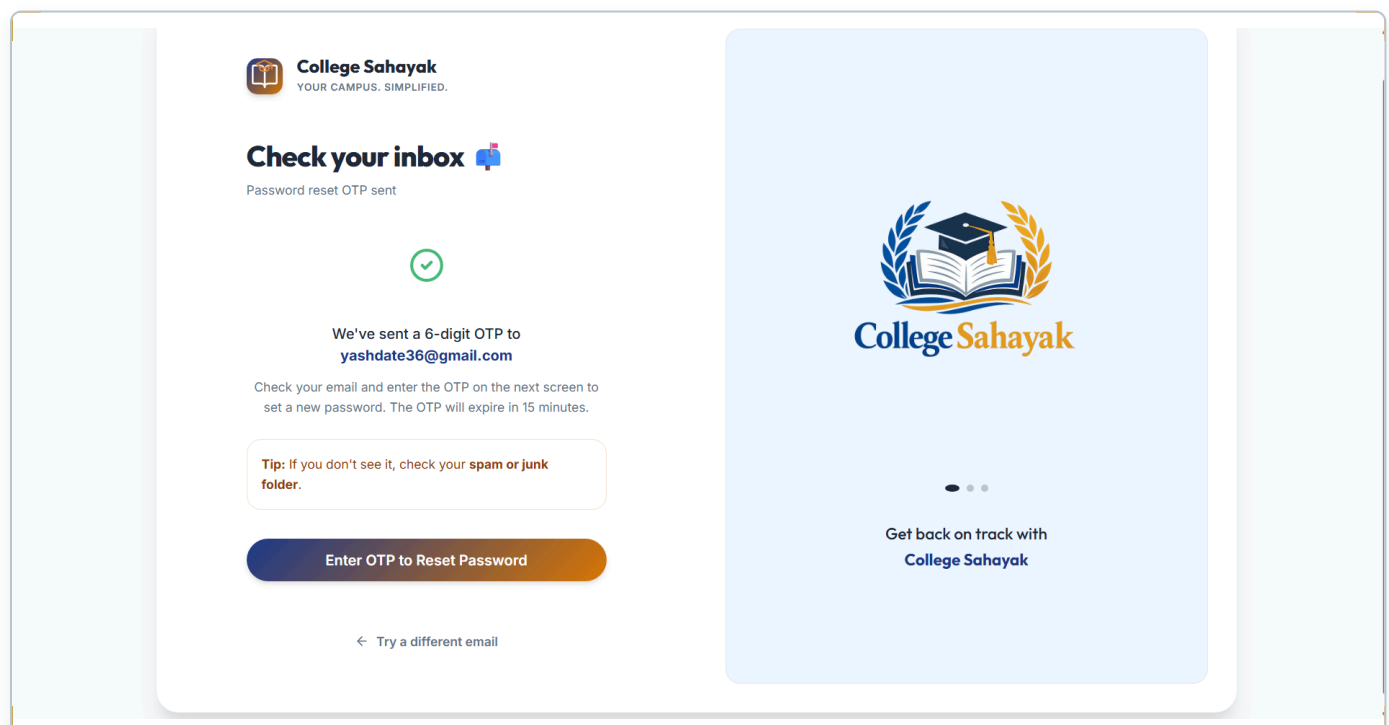


Fig. 3.4 — Forgot & Reset Password Flow

INTERFACE SHOWCASE

Resource Hub & Lab Manuals

Centralized study material repository organized into 10 categories with filtering for branch, semester, and subject.

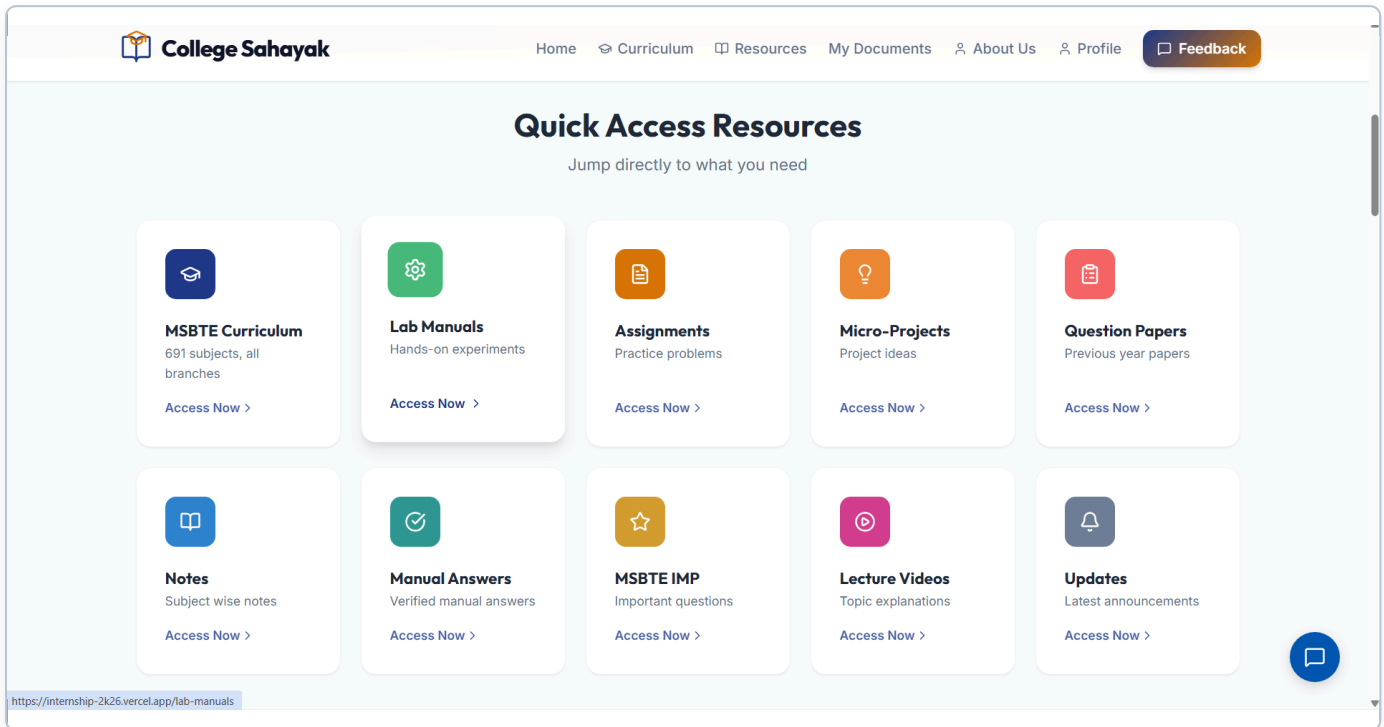


Fig. 3.5 — Materials Hub (10 Resource Categories)

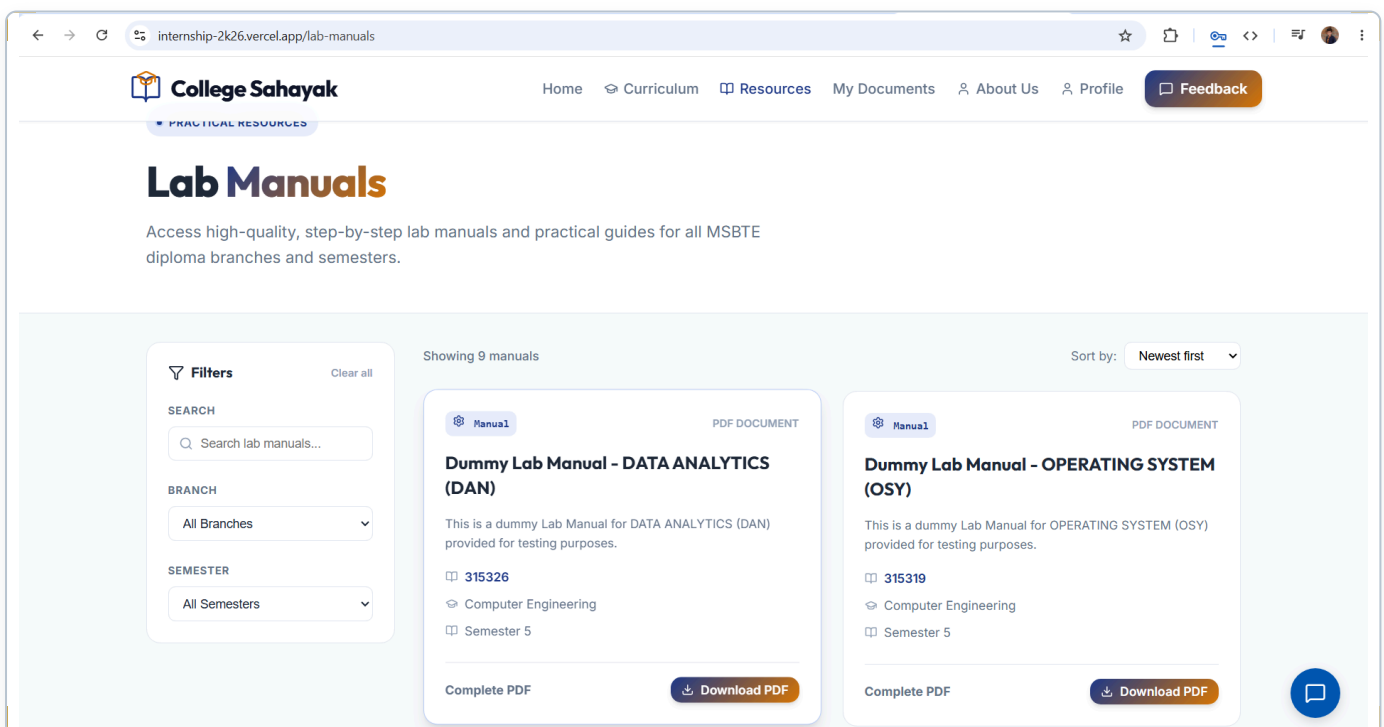


Fig. 3.6 — MSBTE Lab Manuals & Answer Sheets Page

INTERFACE SHOWCASE

User Profile & Document Upload

Student profile management dashboard and crowd-sourced document contribution module ("My Documents").

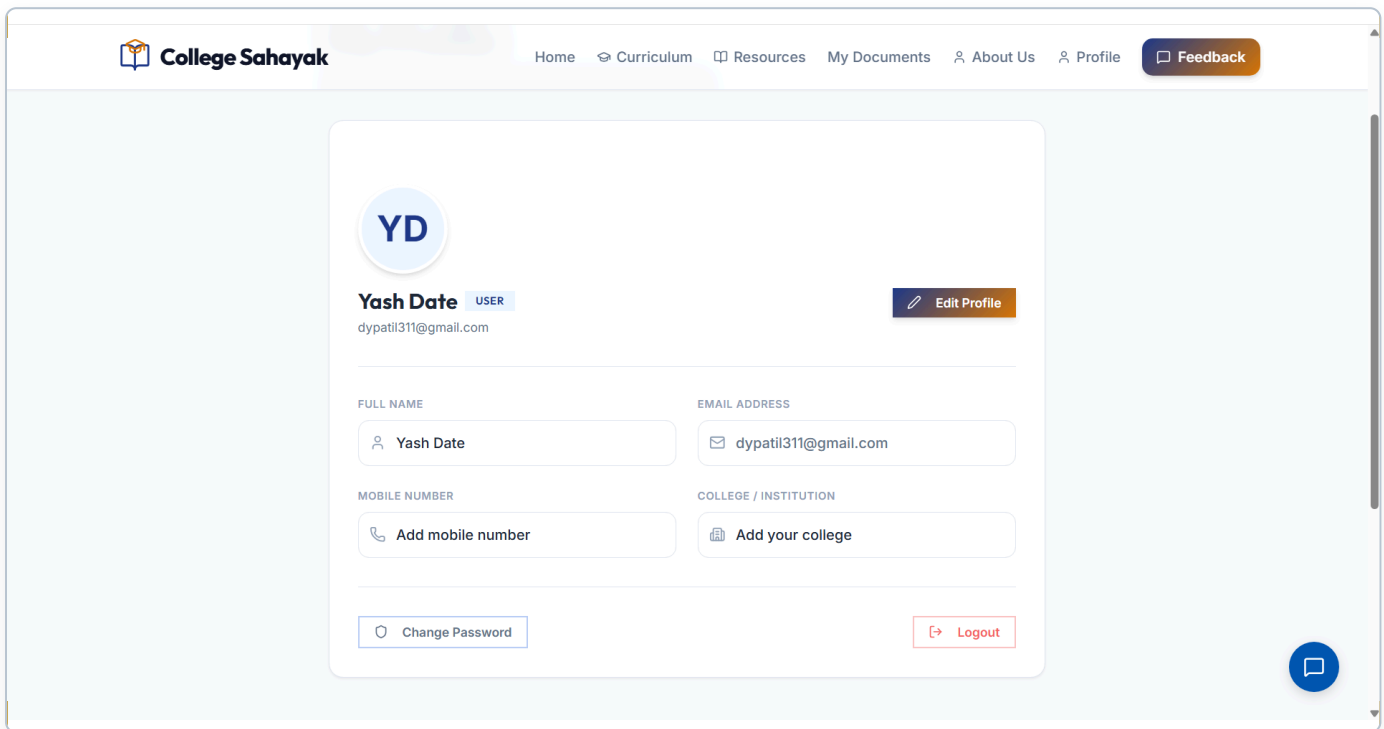


Fig. 3.7 — Student Profile & Account Details

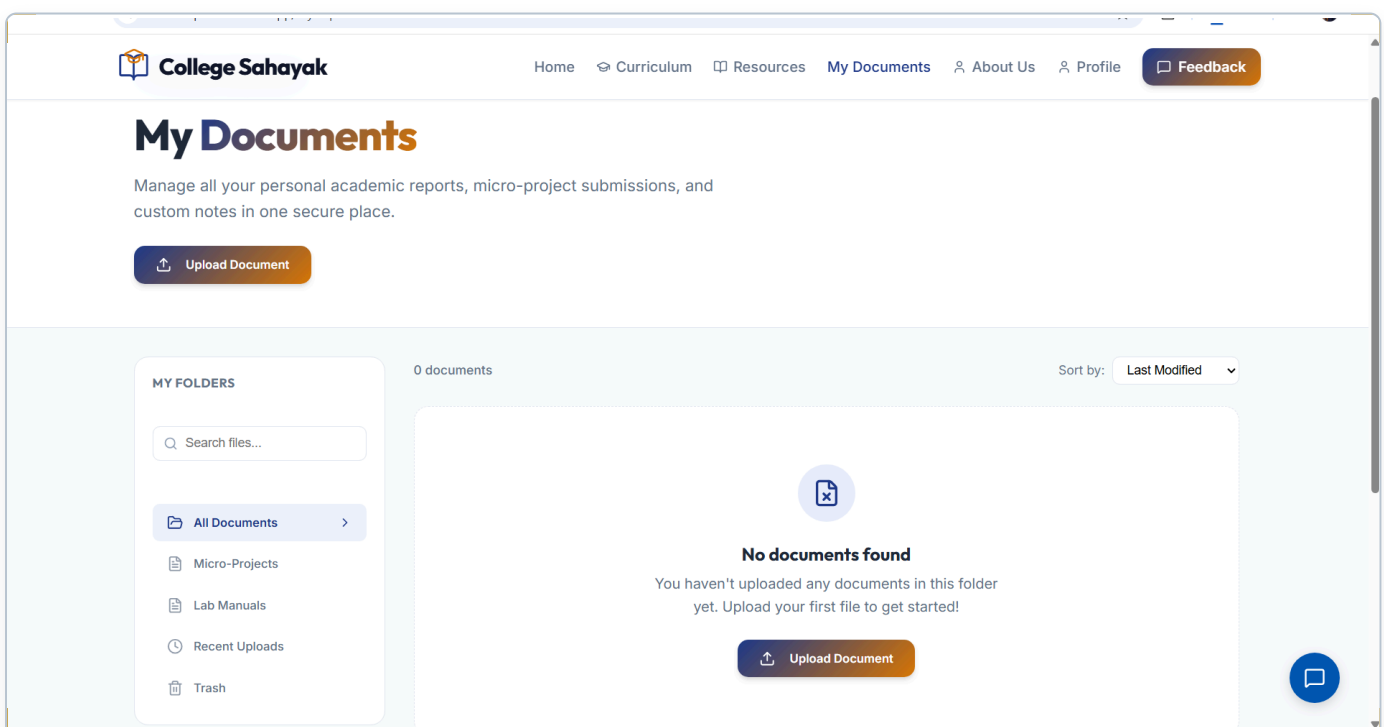


Fig. 3.8 — My Documents & Study Material Submission Portal

INTERFACE SHOWCASE

Admin Panel Dashboard

Role-protected administrative dashboard accessible only to admin users. Features real-time statistics, material management with full CRUD operations, user management, and feedback review — all within a unified control panel.

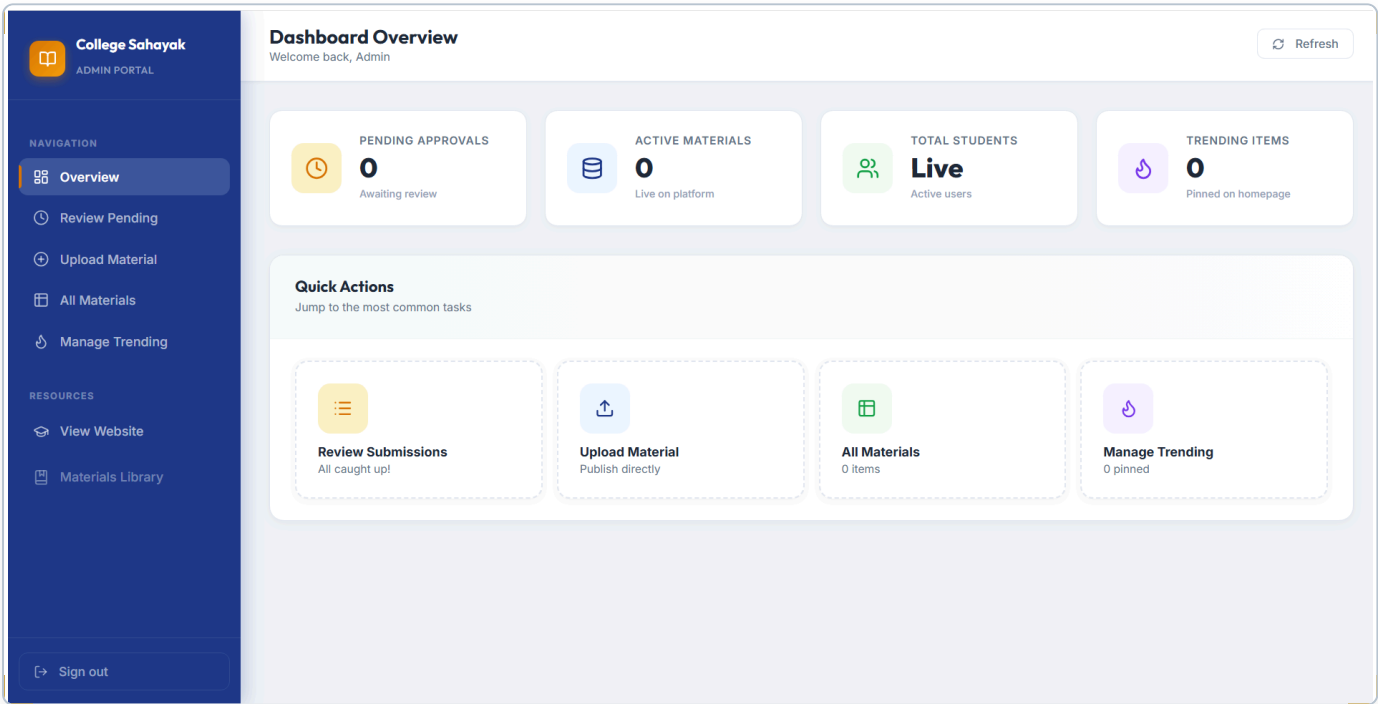


Fig. 3.9 — Admin Panel Dashboard with Material Management & Analytics

INTERFACE SHOWCASE

AI Chatbot & Deployment

An integrated AI chatbot powered by the Google Gemini API (College Mitra), providing contextual study assistance to students — answering academic queries, explaining concepts, and guiding them to relevant resources within the platform.

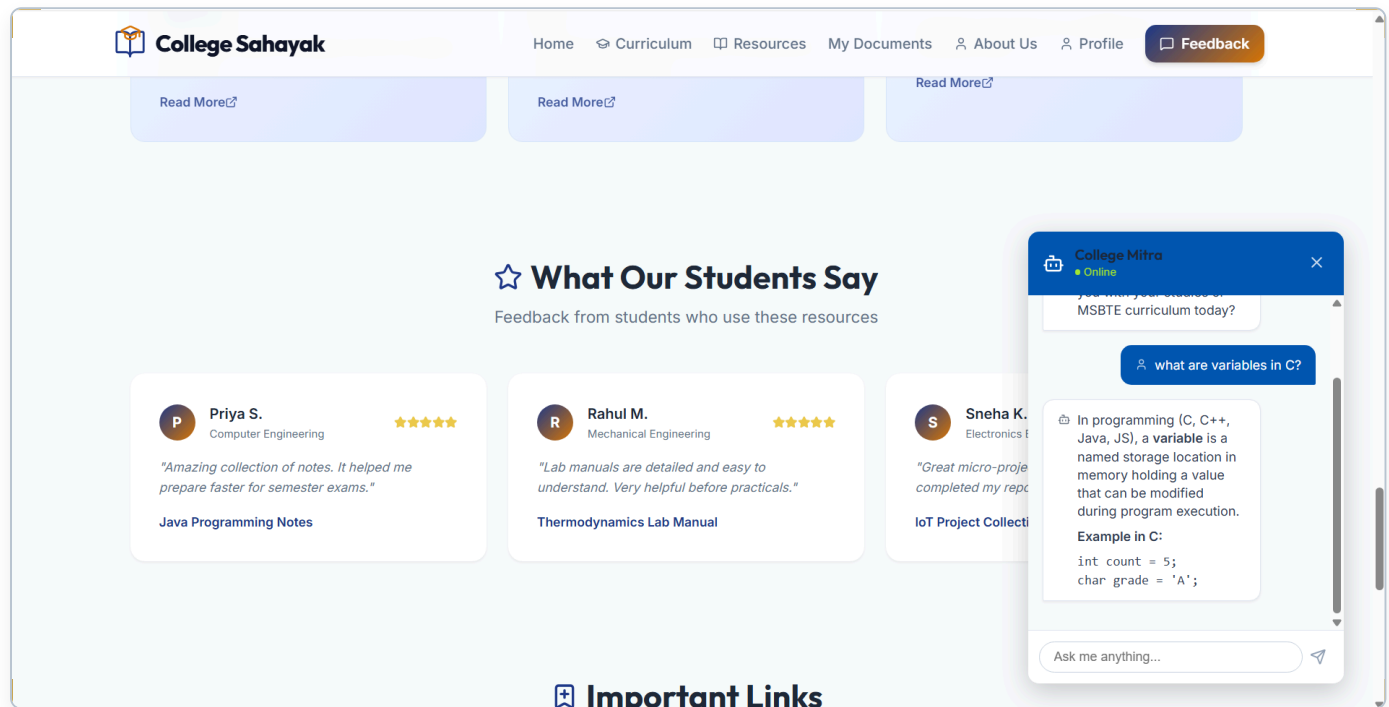


Fig. 3.10 — College Mitra AI Chatbot (Gemini API) with Markdown rendering

AI Chatbot Highlights

- Natural-language conversational interface embedded directly into the application.
- Responses rendered with full Markdown support including code blocks, tables, and lists.
- Backend proxy architecture keeps the Gemini API key secure on the server side.
- Designed to assist students with academic queries, reducing support overhead.

Deployment Details

- **Frontend:** Deployed on Vercel with automatic GitHub CI/CD pipeline.
- **Backend:** Deployed on Render with managed environment variables and auto-scaling.
- **Database:** Hosted on Supabase cloud with PostgreSQL and REST API access.

END OF PORTFOLIO

Thank You

This document summarizes the three internship projects completed at Somayu Infotech — Amazon : The Shopping King e-commerce platform, Power BI analytics dashboard for Hospital ER data, and College Sahayak full-stack web application with AI integration. Each project demonstrates progressive skill development from front-end fundamentals to data analytics and production-grade full-stack engineering.

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